Battery module thermal uniformity in EVs

"is the uneven heating of the battery modules a problem in EVs?" *Generated by LeapSpace*

Yes — **uneven heating inside EV battery modules is a real problem**, because temperature non-uniformity can cause some cells or regions to overheat while others stay cooler, which hurts performance, accelerates degradation, and increases safety risk [113] [114] [115] [116] [117].

Why it matters

Effect of uneven heating	What the evidence shows
Faster degradation	Non-uniform temperature distribution leads to uneven aging, higher resistance growth, and faster capacity fade across cells in a module or pack [116] [117] [118].
Current imbalance	Thermal gradients can drive uneven current sharing in parallel-connected cells, which further amplifies stress on warmer cells [117] [119] [120] [121].
Lower safety margin	Local hot spots raise the risk of thermal runaway and can help propagation within a module or pack [113] [122] [123] [124] [125].
Reduced usable performance	Uneven heating affects battery efficiency, usable energy, and pack-level performance, especially during fast charging or high discharge [126] [127] [128].

What the studies specifically show

- A module with non-uniform temperature can have one area overheat while another remains too cool, and that temperature spread can affect overall performance and even lead to failure [113].
- Experimental and simulation studies repeatedly link thermal gradients to **uneven aging**, **capacity loss**, and **resistance increase** [116] [117] [118].
- In parallel-connected packs, thermal gradients and uneven heat dissipation can create **current inhomogeneity**, which worsens degradation and safety [119] [120] [129] [130].
- Several studies show that keeping temperature variation small improves pack life; for example, reducing module temperature differences to within about **5 °C** is treated as an important design target [114] [125] [131] [132] [133].

Bottom line

Uneven heating is not just a minor inefficiency — it is a core battery-pack design issue because it can trigger imbalance, speed up aging, and raise the likelihood of thermal runaway, especially under fast charging, high load, or poor cooling conditions [113] [116] [117] [125] [127].

Confidence level: **High** — the retrieved evidence is consistent across experiments, simulations, and reviews that thermal non-uniformity harms EV battery performance, durability, and safety [113] [116] [117] [126].

References

113. Guo, Z., Zhou, Z., Li, X., (...), Li, C. (2025) Eutectic hydrated salt composite phase change material for enhancing thermal safety of batteries. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2025.127262

114. Chen, M., Gong, Y., Zhao, L., Chen, Y. (2024) Phase change material with outstanding thermal stability and mechanical strength for battery thermal management. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2024.114565

115. Dubey, D., Mishra, A., Ghosh, S., (...), Pandey, R. (2023) Geometry-influenced cooling performance of lithium-ion battery. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2023.120723

116. Rüwald, T., Marongiu, A., Chahardahcherik, D., (...), Figgemeier, E. (2024) Experimental investigation of cell degradation in packs of parallel-connected cells under different temperature distributions. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2024.114427

117. Liu, X., Ai, W., Naylor Marlow, M., (...), Wu, B. (2019) The effect of cell-to-cell variations and thermal gradients on the performance and degradation of lithium-ion battery packs. *Applied Energy*. https://doi.org/10.1016/j.apenergy.2019.04.108

118. Mussa, A.S., Smith, A.J., Trippetta, G.M., (...), Lindström, R.W. (2025) Performance and aging of a lithium-ion battery in a non-uniform temperature distribution condition. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2025.115869

119. Roehrer, F., Jocher, P., Rehm, M., (...), Jossen, A. (2025) Real-time replication of battery cells using digital twins: A novel experimental methodology to measure current inhomogeneity in n-parallel configurations. *Journal of Power Sources*. https://doi.org/10.1016/j.jpowsour.2025.236573

120. Liang, J., Gan, Y., Tan, M., Li, Y. (2020) Multilayer electrochemical-thermal coupled modeling of unbalanced discharging in a serially connected lithium-ion battery module. *Energy*. https://doi.org/10.1016/j.energy.2020.118429

121. Huang, H.-H., Chen, H.-Y., Liao, K.-C., (...), Tien, J.-Y. (2017) Thermal-electrochemical coupled simulations for cell-to-cell imbalances in lithium-iron-phosphate based battery packs. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2017.05.105

122. Jia, S., Li, N., Shuang, L., (...), Xu, X. (2025) Numerical simulation and experimental study of a novel emergency cooling system for thermal runaway propagation of power battery module. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2025.126344

123. Zhang, Y., Jiang, S., Zhang, Q., (...), Zhang, Y. (2026) Design and performance analysis of a multi-stage phase change material-based thermal management system for lithium-ion batteries. *Journal of Power Sources*. https://doi.org/10.1016/j.jpowsour.2026.239551

124. Shi, J., Zhang, Z., Zhang, M. (2026) Thermal Runaway Simulation and Fire Risk Assessment of Electric Vehicle Power Battery Packs. *Fire*. https://doi.org/10.3390/fire9070313

125. Nagmule, Siddharth A., Salunkhe, Pramod B. (2023) Thermal management of lithium-ion battery module using the phase change material. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*. https://doi.org/10.1177/09544062231191047

126. Nandagopal, K., Sok, R., Kishida, K., (...), Kusaka, J. (2024) Measured Thermal Performances at Brick and Module Levels in a Battery Pack of a Mid-Size Electric Vehicle under WLTC and FTP Cycles. *SAE Technical Papers*. https://doi.org/10.4271/2024-01-2673

127. Allgäuer, C., Huber, J., Gamra, K.A., (...), Lienkamp, M. (2026) Model-based fast charging of lithium-ion batteries: Impact of thermal gradients on the degradation of parallel-connected cells. *eTransportation*. https://doi.org/10.1016/j.etran.2025.100519

128. Hong, S., Li, X., Zeng, W., (...), Wang, M. (2026) Expanding the lithium plating boundary via a chemical vapor deposition–derived silicon–carbon anode for ultra-fast-charging large cylindrical cells. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2026.122766

129. Li, C., Chang, L., Ma, C., Duan, B. (2026) Cell-to-cell inconsistency in parallel battery packs under uneven heat dissipation: Experimental and simulation analysis. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2025.120063

130. Hosseinzadeh, E., Arias, S., Krishna, M., (...), Marco, J. (2021) Quantifying cell-to-cell variations of a parallel battery module for different pack configurations. *Applied Energy*. https://doi.org/10.1016/j.apenergy.2020.115859

131. Hürdoğan, E., Coşanay, H., Şenocak, Ş.M., (...), Farsak, M. (2026) A comprehensive numerical investigation on the thermal performance of forced air-cooled cylindrical battery packs. *Journal of Thermal Analysis and Calorimetry*. https://doi.org/10.1007/s10973-026-16304-7

132. Sun, Z., Guo, Y., Zhang, C., (...), Wang, C. (2023) Experimental study of battery passive thermal management system using copper foam-based phase change materials. *International Journal of Thermofluids*. https://doi.org/10.1016/j.ijft.2022.100255

133. Wankhede, S., More, K., Kamble, L., Thorat, P. (2024) Experimental investigation on hybrid cooled lithium-ion battery pack with 3S4P cell configuration using OM 48 as phase change material and heat pipe. *Energy Storage*. https://doi.org/10.1002/est2.496

Addressing uneven battery module heating

"in the current thermal management strategies, how is this uneven heating addressed?" *Generated by LeapSpace*

How uneven heating is addressed

Uneven heating in EV battery packs is mainly addressed by **making coolant delivery more uniform, spreading heat more evenly, buffering hot spots, and actively coordinating cooling with pack layout and control** [1] [79] [134] [135]. In practice, the evidence points to a few recurring design moves:

Strategy	How it reduces uneven heating	Evidence
Optimize coolant flow paths	Adjusting inlet/outlet layout, alternating or counter-flow channels, and flow direction helps reduce inlet-to-outlet temperature differences and improves pack uniformity [136] [137] [138] [139] [140] [141]	Long channels can create a large coolant temperature rise along the flow path, so redesigned flow arrangements are used to keep temperatures more even [136] [137].
Use PCM to buffer local hot spots	Phase change materials absorb transient heat spikes and slow temperature rise, which improves temperature equalization across cells [142] [143] [144] [145] [146] [147]	PCM-based systems are repeatedly reported to improve uniformity and lower peak temperature, especially when paired with other methods [143] [145] [147].
Combine PCM with active cooling	Hybrid PCM + liquid cooling or PCM + heat pipe systems handle both local buffering and continuous heat removal, reducing uneven heating better than PCM alone [141] [148] [149] [150] [151] [152]	Several hybrid designs achieve lower maximum temperature and tighter temperature spread than single-technique systems [150] [151] [152].
Improve heat spreading with heat pipes or plates	Heat pipes, cold plates, splitter plates, and guide vanes move heat away from hotspots and flatten gradients across the module [134] [153] [154] [155] [156] [157]	Heat pipes are especially used to transport heat from cells to a more uniform rejection surface, while plate geometry helps even out contact-side temperatures [153] [155] [156].
Tune pack geometry and channel structure	Cell spacing, channel width, staggered layouts, microchannels, and segmented cold plates are optimized to balance heat removal and avoid local overcooling/undercooling [158] [159] [160] [161] [162] [163]	Design optimization can reduce localized hot spots while keeping pressure drop and pumping losses acceptable [161] [162] [163].

Strategy	How it reduces uneven heating	Evidence
Use smarter control and sensing	Model-based control, predictive optimization, and coordinated battery-cabin thermal control help direct cooling where and when it is needed [135] [164] [165] [166] [167]	The literature increasingly treats thermal uniformity as a control problem, not only a hardware problem [135] [164] [165].

What the evidence says in plain terms

- **Liquid cooling** is the main baseline because flow arrangement and channel design can directly improve temperature uniformity across the module [137] [139] [163] [168].
- **PCM systems** help because they absorb heat locally, but they work best when paired with a way to remove the stored heat later [146] [169] [170].
- **Hybrid systems** are the strongest response to uneven heating because they combine heat spreading, heat storage, and active heat rejection [148] [149] [150] [151] [157].
- **Geometry matters as much as cooling power**: better coolant distribution, cross-flow or counter-flow layouts, and segmented or optimized cold plates can cut temperature spread without simply increasing flow rate [140] [141] [162] [167].
- **Uneven heating is also managed by avoiding cabin-battery conflicts** in integrated systems, since the battery branch may need higher flow than the cabin branch to preserve uniformity [171].

Bottom line

Uneven heating is addressed by **engineering the thermal path, not just adding more cooling**: the current direction is to **redistribute coolant more intelligently, add PCM or heat-pipe buffering, and use hybrid, model-driven control so the pack stays both cooler and more uniform** [1] [137] [150] [151] [166] [167].

Confidence level: **High** — the retrieved studies consistently converge on flow optimization, PCM buffering, hybrid architectures, and control-aware thermal design as the main ways to reduce module-level temperature non-uniformity [137] [141] [150] [151] [166].

References

1. Liu, H., Wei, Z., He, W., Zhao, J. (2017) Thermal issues about Li-ion batteries and recent progress in battery thermal management systems: A review. *Energy Conversion and Management*. https://doi.org/10.1016/j.enconman.2017.08.016

79. Amer, M.M., Shouman, M.A., Salem, M.S., (...), Hamed, A.M. (2024) Advances in thermal management systems for Li-Ion batteries: A review. *Thermal Science and Engineering Progress*. https://doi.org/10.1016/j.tsep.2024.102714

134. Ismailov, K., Adair, D., Massalin, Y., Bakenov, Z. (2017) On using splitter plates and flow guide-vanes for battery module cooling. *Heat and Mass Transfer/Waerme- und Stoffuebertragung*. https://doi.org/10.1007/s00231-016-1794-5

135. El-Sharkawy, A., Asar, M., Serpento, S., Sheta, M. (2026) Modeling Non-Uniform Temperature Distribution in Lithium-Ion Batteries. *SAE Technical Papers*. https://doi.org/10.4271/2026-01-0120

136. Cao, J., Ling, Z., Fang, X., Zhang, Z. (2020) Delayed liquid cooling strategy with phase change material to achieve high temperature uniformity of Li-ion battery under high-rate discharge. *Journal of Power Sources*. https://doi.org/10.1016/j.jpowsour.2019.227673

137. Yue, Q.L., He, C.X., Wu, M.C., (...), Zhao, T.S. (2022) Pack-level modeling of a liquid cooling system for power batteries in electric vehicles. *International Journal of Heat and Mass Transfer*. https://doi.org/10.1016/j.ijheatmasstransfer.2022.122946

138. Xie, Y., Li, L., Xu, Q., (...), Zheng, K. (2024) Effects of module size on the heat dissipation performance of the thermal management system for the battery module with cylindrical cells. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2023.121958

139. Keyinci, S., Ilinčić, P., Tosun, E., (...), Özcanlı, M. (2025) Design and performance analysis of a curved channel liquid cooling system for battery thermal management. *Thermal Science and Engineering Progress*. https://doi.org/10.1016/j.tsep.2025.103690

140. Wu, Y., Yan, Y., Zhang, C., (...), Huang, J. (2026) Numerical investigation of externally wrapped circulation cooling channels for cylindrical lithium-ion battery thermal management: Enhancing thermal uniformity and energy efficiency. *International Journal of Heat and Mass Transfer*. https://doi.org/10.1016/j.ijheatmasstransfer.2025.128281

141. Shen, J., Chen, X., Xu, X., (...), Zhou, F. (2024) Thermal performance of a hybrid cooling plate integrated with microchannels and PCM. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2023.121917

142. Bai, F., Chen, M., Song, W., (...), Ding, Y. (2019) Investigation of thermal management for lithium-ion pouch battery module based on phase change slurry and mini channel cooling plate. *Energy*. https://doi.org/10.1016/j.energy.2018.10.137

143. Xu, W., Yang, W., Su, J., (...), Li, X. (2024) Diatom-based biomass composites phase change materials with high thermal conductivity for battery thermal management. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2024.112737

144. Nagmule, S.A., Salunkhe, P.B. (2023) Thermal management of lithium-ion battery module using the phase change material. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*. https://doi.org/10.1177/09544062231191047

145. Li, C., Ding, Y., Zhou, Z., (...), Hu, H. (2023) Parameter optimization and sensitivity analysis of a Lithium-ion battery thermal management system integrated with composite phase change material. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2023.120530

146. Wang, Y., Wang, Z., Min, H., (...), Li, Q. (2021) Performance investigation of a passive battery thermal management system applied with phase change material. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2021.102279

147. Sun, J., Dan, D., Wei, M., (...), Wright, E. (2023) Pack-Level Modeling and Thermal Analysis of a Battery Thermal Management System with Phase Change Materials and Liquid Cooling. *Energies*. https://doi.org/10.3390/en16155815

148. Xu, Q., Xie, Y., Huang, H., (...), Zheng, K. (2025) Parametric studies and design recommendations for a novel hybrid battery thermal management system with ultrathin PCM layers between liquid cooling channels and batteries. *Energy*. https://doi.org/10.1016/j.energy.2025.138622

149. Xin, Q., Du, C., Yan, F., (...), Li, W. (2024) Performance optimization of a prismatic battery module integrated with PCM, FHP and liquid cooling for rapid discharge operation and abnormal heating conditions. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2024.113808

150. Rong, L., Bai, X.-S., Li, J.-C., (...), Yang, W.-W. (2024) Design and optimization of a hybrid cooling configuration combining PCM and liquid cooling for Li-ion battery using data-based response surface approximation model. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2024.122844

151. Xin, Q., Yang, T., Zhang, H., (...), Xiao, J. (2023) Experimental and numerical study of lithium-ion battery thermal management system using composite phase change material and liquid cooling. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2023.108003

152. Liu, Z., Xu, G., Xia, Y., Tian, S. (2023) Numerical study of thermal management of pouch lithium-ion battery based on composite liquid-cooled phase change materials with honeycomb structure. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2023.108001

153. Jose, J., Sutheesh, P.M., Bandaru, B., Hotta, T.K. (2026) Development of a heat pipe-based thermal management system for the temperature control of prismatic Lithium-ion batteries: A numerical perspective. *International Communications in Heat and Mass Transfer*. https://doi.org/10.1016/j.icheatmasstransfer.2025.110406

154. Singh, R., Lapp, G., Velardo, J., (...), Busse, K. (2021) Battery cooling options in electric vehicles with heat pipes. *Frontiers in Heat and Mass Transfer*. https://doi.org/10.5098/hmt.16.2

155. Wu, M.-S., Liu, K.H., Wang, Y.-Y., Wan, C.-C. (2002) Heat dissipation design for lithium-ion batteries. *Journal of Power Sources*. https://doi.org/10.1016/S0378-7753(02)00048-4

156. Lai, Y., Ding, J., Liu, L. (2025) Numerical investigation on the cooling performance of lithium-ion battery using liquid cooled-plate with integrated grooves and secondary microchannel structures. *International Journal of Thermal Sciences*. https://doi.org/10.1016/j.ijthermalsci.2025.110094

157. Wang, Y. (2024) Liquid cooling (LC), phase change material cooling (PCMC) and heat pipe cooling (HPC): Comparison and integration of three technologies for thermal management of electric vehicle batteries. *AIP Conference Proceedings*. https://doi.org/10.1063/5.0216295

158. Abdulhaleem Abdulqader, A., Jaffal, H.M. (2024) Comparative investigation on heat transfer augmentation in a liquid cooling plate for rectangular Li-ion battery thermal management. *Results in Engineering*. https://doi.org/10.1016/j.rineng.2024.102913

159. Asadi-Ghajarlu, Z., Ghasemloo, A., Sadigh Saatloo, S., Moeini-Aghtaie, M. (2026) Simulation and optimal design of thermal management systems for EV battery packs. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2026.131292

160. Le, T.D., Lee, M.-Y. (2026) Optimization of thermal performance characteristics for 21,700 lithium-ion battery pack utilizing single-phase immersion cooling: A multi-parameter numerical investigation. *International Communications in Heat and Mass Transfer*. https://doi.org/10.1016/j.icheatmasstransfer.2026.110651

161. Meng, J.-H., Ma, C.-Y., Liu, Y., (...), Lu, G. (2025) Artificial neural network (ANN) based prediction and optimization of battery thermal management for non-uniform flow channel design. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2025.115404

162. Zhang, J., Wang, D., Wang, Y., (...), Li, W. (2026) Segmented lattice liquid cooling structure for high-efficiency thermal management of large-scale lithium-ion battery packs. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2026.130623

163. Bianco, N., Fragnito, A., Iasiello, M., (...), Piccirillo, F. (2026) Numerical evaluation of different cooling strategies for cylindrical battery packs: from air natural convection to topology-optimized cold plates. *International Journal of Thermal Sciences*. https://doi.org/10.1016/j.ijthermalsci.2026.110773

164. Peprah, G.K., Huang, Y., Wik, T., (...), Zou, C. (2026) Thermal Modeling of Battery Cells for Optimal Tab and Surface Cooling Control. *IEEE Transactions on Control Systems Technology*. https://doi.org/10.1109/TCST.2026.3675726

165. Dong, T., Huang, L., Liu, J., (...), Cao, W. (2025) Advances and challenges in obtaining internal temperature for lithium-ion batteries. *Journal of Power Sources*. https://doi.org/10.1016/j.jpowsour.2025.237571

166. Liu, S., Xu, L., Zhang, J., (...), Li, H. (2025) Optimized design of dual-circuit dynamic coordinated control for liquid cooling in large-capacity energy storage lithium battery packs. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2025.118311

167. Kalkan, O. (2026) GMDH and NSGA-III based design optimization of bio-inspired cooling systems for prismatic LFP batteries. *Journal of Power Sources*. https://doi.org/10.1016/j.jpowsour.2026.241256

168. Akbarzadeh, M., Jaguemont, J., Kalogiannis, T., (...), Berecibar, M. (2021) A novel liquid cooling plate concept for thermal management of lithium-ion batteries in electric vehicles. *Energy Conversion and Management*. https://doi.org/10.1016/j.enconman.2021.113862

169. Li, J., Tang, A., Shao, X., (...), Xia, D. (2022) Experimental evaluation of heat conduction enhancement and lithium-ion battery cooling performance based on h-BN-based composite phase change materials. *International Journal of Heat and Mass Transfer*. https://doi.org/10.1016/j.ijheatmasstransfer.2021.122487

170. Rawat, S.S., Kumar, R. (2024) Techno-economic analysis of cooling technologies used in electric vehicle battery thermal management: A review. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2024.114135

171. Jia, F., Yin, X., Wang, A., (...), Wang, X. (2026) Dynamic trade-off balancing and energy savings in battery-cabin coupled CO2 EVTMS via semi-series refrigerant direct cooling architecture. *Energy Conversion and Management*. https://doi.org/10.1016/j.enconman.2025.120977

Limitations of battery thermal management systems

"what are the limitations the existing systems are facing in cooling the battery evenly?" *Generated by LeapSpace*

Direct answer

Existing battery thermal management systems struggle to cool packs **evenly** because they face a mix of **low heat-removal capacity, poor temperature uniformity, flow-induced hot spots, and trade-offs between cooling performance and system complexity** [1] [2] [4] [5] [60] [81] [172] [174]. In short, many systems can reduce average temperature, but they still have trouble keeping the **temperature difference across cells/modules small** under fast charging, high load, or harsh ambient conditions [81] [173] [176] [201].

Main limitations affecting even cooling

Limitation	How it affects even cooling	Evidence
Air cooling has weak capacity	Air systems often cannot remove heat fast enough, so some cells stay hotter than others, especially under high load [1] [5] [6] [175]	Low heat-removal capacity and weak temperature uniformity are recurring issues [1] [5] [6] [175]
Liquid cooling can still be non-uniform	Liquid systems perform better overall, but channel layout, inlet-outlet effects, and flow distribution can still create temperature gradients across the pack [60] [137] [138] [139] [140] [163] [168]	Geometry and flow-path design strongly affect uniformity [137] [139] [140] [163] [168]
PCM mainly buffers, not continuously removes heat	Phase change materials can absorb local spikes, but they have low thermal conductivity and limited secondary heat rejection, so hotspots can persist once the PCM saturates [13] [180] [181] [182] [183]	PCM is effective for buffering, but not as a standalone solution for sustained high loads [172] [179]
Heat pipes still need a heat sink	Heat pipes spread heat well, but they do not eliminate the need for an external rejection path, so uneven cooling can remain if the downstream sink is weak [153] [155] [184]	Performance depends on arrangement and startup behavior [153] [155] [184]

Limitation	How it affects even cooling	Evidence
Thermoelectric systems are less efficient	They can control temperature precisely, but their energy efficiency is often weaker than fluid-based systems, limiting their ability to serve as the only cooling method [185] [186] [187] [188]	Better suited to hybrid use than standalone deployment [187] [188]
Hybrid systems are effective but complex	Combining PCM, liquid cooling, heat pipes, or thermoelectrics improves uniformity, but adds cost, weight, integration burden, and control complexity [174] [179] [185] [189] [190] [191]	The main challenge is scaling strong prototypes into practical vehicle systems [178] [192] [193]

What this means in practice

- **Uneven coolant distribution** is a major reason hot spots persist, so better cooling is not just about more power but about better flow design [136] [137] [140] [161] [162].
- **Thermal uniformity is constrained by packaging and system integration**, not just by the cooling hardware itself [177] [178] [192] [193] [202].
- **Fast charging and extreme weather expose these weaknesses most clearly**, because many systems cannot keep both peak temperature and temperature spread within safe limits [81] [173] [176] [197] [198] [199] [200] [201].
- **The central trade-off** is that systems that cool more evenly often become heavier, costlier, or more complex, while simpler systems struggle with hot spots and gradients [81] [172] [174] [179].

Bottom line

The biggest limitation is **not the absence of cooling technologies**, but the difficulty of achieving **uniform heat removal across the whole pack** without unacceptable penalties in energy use, weight, cost, or complexity [81] [172] [174] [179]. Current systems still struggle most with **flow non-uniformity, low passive heat rejection, and maintaining even cooling under fast charging or extreme conditions** [1] [5] [60] [81] [173] [176].

Confidence level: **High** — the cited reviews and studies consistently identify the same bottlenecks across air, liquid, PCM, heat-pipe, thermoelectric, and hybrid battery thermal management systems [1] [2] [4] [60] [81] [172] [174].

References

1. Liu, H., Wei, Z., He, W., Zhao, J. (2017) Thermal issues about Li-ion batteries and recent progress in battery thermal management systems: A review. *Energy Conversion and Management*. https://doi.org/10.1016/j.enconman.2017.08.016

2. Kim, J., Oh, J., Lee, H. (2019) Review on battery thermal management system for electric vehicles. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2018.12.020

4. An, Z., Jia, L., Ding, Y., (...), Li, X. (2017) A review on lithium-ion power battery thermal management technologies and thermal safety. *Journal of Thermal Science*. https://doi.org/10.1007/s11630-017-0955-2

5. Zhao, G., Wang, X., Negnevitsky, M., Zhang, H. (2021) A review of air-cooling battery thermal management systems for electric and hybrid electric vehicles. *Journal of Power Sources*. https://doi.org/10.1016/j.jpowsour.2021.230001

6. Yang, N., Zhang, X., Li, G., Dong, D. (2015) Assessment of the forced air-cooling performance for cylindrical lithium-ion battery packs: A comparative analysis between aligned and staggered cell arrangements. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2015.01.049

13. Rao, Z., Wang, S. (2011) A review of power battery thermal energy management. *Renewable and Sustainable Energy Reviews*. https://doi.org/10.1016/j.rser.2011.07.096

60. Wu, W., Wang, S., Wu, W., (...), Lai, Y. (2019) A critical review of battery thermal performance and liquid based battery thermal management. *Energy Conversion and Management*. https://doi.org/10.1016/j.enconman.2018.12.051

81. Yerumbu, N. (2026) Recent advances and challenges in electric vehicle Li-ion battery cooling systems. *Next Energy*. https://doi.org/10.1016/j.nxener.2026.100720

136. Cao, J., Ling, Z., Fang, X., Zhang, Z. (2020) Delayed liquid cooling strategy with phase change material to achieve high temperature uniformity of Li-ion battery under high-rate discharge. *Journal of Power Sources*. https://doi.org/10.1016/j.jpowsour.2019.227673

137. Yue, Q.L., He, C.X., Wu, M.C., (...), Zhao, T.S. (2022) Pack-level modeling of a liquid cooling system for power batteries in electric vehicles. *International Journal of Heat and Mass Transfer*. https://doi.org/10.1016/j.ijheatmasstransfer.2022.122946

138. Xie, Y., Li, L., Xu, Q., (...), Zheng, K. (2024) Effects of module size on the heat dissipation performance of the thermal management system for the battery module with cylindrical cells. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2023.121958

139. Keyinci, S., Ilinčić, P., Tosun, E., (...), Özcanlı, M. (2025) Design and performance analysis of a curved channel liquid cooling system for battery thermal management. *Thermal Science and Engineering Progress*. https://doi.org/10.1016/j.tsep.2025.103690

140. Wu, Y., Yan, Y., Zhang, C., (...), Huang, J. (2026) Numerical investigation of externally wrapped circulation cooling channels for cylindrical lithium-ion battery thermal management: Enhancing thermal uniformity and energy efficiency. *International Journal of Heat and Mass Transfer*. https://doi.org/10.1016/j.ijheatmasstransfer.2025.128281

153. Jose, J., Sutheesh, P.M., Bandaru, B., Hotta, T.K. (2026) Development of a heat pipe-based thermal management system for the temperature control of prismatic Lithium-ion batteries: A numerical perspective. *International Communications in Heat and Mass Transfer*. https://doi.org/10.1016/j.icheatmasstransfer.2025.110406

155. Wu, M.-S., Liu, K.H., Wang, Y.-Y., Wan, C.-C. (2002) Heat dissipation design for lithium-ion batteries. *Journal of Power Sources*. https://doi.org/10.1016/S0378-7753(02)00048-4

161. Meng, J.-H., Ma, C.-Y., Liu, Y., (...), Lu, G. (2025) Artificial neural network (ANN) based prediction and optimization of battery thermal management for non-uniform flow channel design. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2025.115404

162. Zhang, J., Wang, D., Wang, Y., (...), Li, W. (2026) Segmented lattice liquid cooling structure for high-efficiency thermal management of large-scale lithium-ion battery packs. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2026.130623

163. Bianco, N., Fragnito, A., Iasiello, M., (...), Piccirillo, F. (2026) Numerical evaluation of different cooling strategies for cylindrical battery packs: from air natural convection to topology-optimized cold plates. *International Journal of Thermal Sciences*. https://doi.org/10.1016/j.ijthermalsci.2026.110773

168. Akbarzadeh, M., Jaguemont, J., Kalogiannis, T., (...), Berecibar, M. (2021) A novel liquid cooling plate concept for thermal management of lithium-ion batteries in electric vehicles. *Energy Conversion and Management*. https://doi.org/10.1016/j.enconman.2021.113862

172. Wankhede, S., Thorat, P., Shisode, S., (...), Wankhade, R. (2022) A study of different battery thermal management systems for battery pack cooling in electric vehicles. *Heat Transfer*. https://doi.org/10.1002/htj.22653

173. Keyser, M., Pesaran, A., Li, Q., (...), Markel, A. (2017) Enabling fast charging – Battery thermal considerations. *Journal of Power Sources*. https://doi.org/10.1016/j.jpowsour.2017.07.009

174. https://www.scopus.com/pages/publications/85207771033

175. Bhatnagar, P., Mittal, G., Bisht, V., Saxena, A. (2024) CFD modelling of an air cooling battery thermal management system. *AIP Conference Proceedings*. https://doi.org/10.1063/5.0183116

176. Li, T., Wang, Z., Chen, L. (2026) Topology optimization of liquid-cooled plates based on a power-law hybrid interpolation model and performance analysis. *International Journal of Thermal Sciences*. https://doi.org/10.1016/j.ijthermalsci.2026.110815

177. Gharehghani, A., Rabiei, M., Mehranfar, S., (...), Micó, C.M. (2024) Progress in battery thermal management systems technologies for electric vehicles. *Renewable and Sustainable Energy Reviews*. https://doi.org/10.1016/j.rser.2024.114654

178. Rohit, K., Ahirwar, B.K., Verma, A. (2025) Integration of battery thermal management systems (BTMS) into the Indian automotive electric vehicle supply chain: a comprehensive review. *Journal of Thermal Analysis and Calorimetry*. https://doi.org/10.1007/s10973-025-14655-1

179. Rathod, Manish K., Patel, Jay R. (2023) A novel cooling strategy for lithium-ion battery thermal management with phase change material. *Handbook of Thermal Management Systems*. https://doi.org/10.1016/B978-0-443-19017-9.00036-2

180. Gasmelseed, A., AlMallahi, M.N., Elgendi, M. (2026) Enhancement strategies for PCM based battery thermal management systems: A review. *Materials Research Proceedings*. https://doi.org/10.21741/9781644904176-18

181. Paul, D., Nandi, A., Bhattacharyya, S., (...), Biswas, N. (2025) A comprehensive review on lithium-ion battery thermal management (BTM) using phase change materials: advances, challenges, and future perspectives. *Journal of Thermal Analysis and Calorimetry*. https://doi.org/10.1007/s10973-025-14260-2

182. Farid, M.M., Auckaili, A.M., Razack, S.A.K., Al-Hallaj, S. (2004) A review on phase change energy storage: Materials and applications. *Energy Conversion and Management*. https://doi.org/10.1016/j.enconman.2003.09.015

183. An, Z., Huang, Z., Mombeki Pea, H.J., (...), Ding, Y. (2026) PCM-based thermal management and fire suppression of lithium-ion batteries: A comprehensive review of current technologies and future directions. *Renewable and Sustainable Energy Reviews*. https://doi.org/10.1016/j.rser.2025.116369

184. Fathoni, A.M., Hendrayanto, P.A., Aliefiansyah, M.S., Putra, N. (2024) Startup characteristics of a water loop heat pipe with dual heat sources for battery thermal management system in electric vehicle. *IOP Conference Series: Earth and Environmental Science*. https://doi.org/10.1088/1755-1315/1372/1/012013

185. An, Z., Liu, H., Gao, W., Zhang, J. (2024) A triple-hybrid battery thermal management system with drop-shaped fin channels for improving weather tolerance. *Energy*. https://doi.org/10.1016/j.energy.2024.132752

186. Rahmani, A., Dibaj, M., Akrami, M. (2024) A study on a battery pack in a hybrid battery thermal management system integrating with thermoelectric cooling. *Case Studies in Thermal Engineering*. https://doi.org/10.1016/j.csite.2024.104856

187. Wu, Z., Luo, D., Zhang, P., (...), Chen, H. (2026) Rapid response of the thermoelectric active battery thermal management system assisted with phase change materials. *Journal of Advanced Research*. https://doi.org/10.1016/j.jare.2026.02.011

188. Pakrouh, R., Hosseini, M.J., Ranjbar, A.A., Rahimi, M. (2023) A novel liquid-based battery thermal management system coupling with phase change material and thermoelectric cooling. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2023.107098

189. Suresh, C., Awasthi, A., Kumar, B., (...), Jeon, Y. (2025) Advances in battery thermal management for electric vehicles: A comprehensive review of hybrid PCM-metal foam and immersion cooling technologies. *Renewable and Sustainable Energy Reviews*. https://doi.org/10.1016/j.rser.2024.115021

190. Dubey, G.P., Mannepalli, K., Sowmya, H.K., (...), Khishe, M. (2026) Hybrid IoT–ML-assisted PCM-based battery thermal management system for lithium-ion battery modules: Experimental investigation and thermal optimization. *Hybrid Advances*. https://doi.org/10.1016/j.hybadv.2026.100681

191. Siricharoenpanich, A., Hamza, H., Naphon, P. (2026) Enhancing heat dissipation in cylindrical battery packs via phase change materials embedded with TiO_2/Fe_3O_4 nanoparticles. *Results in Engineering*. https://doi.org/10.1016/j.rineng.2026.111460

192. Li, X., Wang, R. (2025) Towards integrated thermal management systems in battery electric vehicles: A review. *eTransportation*. https://doi.org/10.1016/j.etran.2025.100396

193. Dhanasekaran, A., Subramanian, Y., Thang Le, M., (...), Azad, A.K. (2026) Battery thermal management systems for electric vehicles: A review of current status, opportunities, and future challenges. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2026.122330

197. Nazlıgül, H., Savrun, M.M., Doğru Mert, B. (2026) Extreme weather impacts on LiFePO4 batteries: Insights from arrhenius-based analysis and charging techniques. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2025.119820

198. Gao, J., Zhang, N., Zhang, T. (2025) Cold waves and electric vehicle adoption: Evidence from Chinese cities. *Transportation Research Part D: Transport and Environment*. https://doi.org/10.1016/j.trd.2025.104748

199. Maher, K., Boumaiza, A. (2025) Thermal challenges in lithium-ion battery technology: Investigating performance and thermal stability. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2025.115396

200. Gou, J., Liu, W. (2019) Feasibility study on a novel 3D vapor chamber used for Li-ion battery thermal management system of electric vehicle. *Applied Thermal Engineering*. https://doi.org/10.1016/j.applthermaleng.2019.02.034

201. Budiman, A.C., Kaleg, S., Sudirja, S., (...), Hapid, A. (2022) Passive thermal management of battery module using paraffin-filled tubes: An experimental investigation. *Engineering Science and Technology, an International Journal*. https://doi.org/10.1016/j.jestch.2021.06.011

202. Rawat, S., Kumar, R. (2025) Battery thermal management systems for electric vehicles: current trends and strategies for sustainable transportation. *Engineering Research Express*. https://doi.org/10.1088/2631-8695/ae06f3